

Zenghong He

zeh012@ucsd.edu | Phone:+1 650-609-0161 | Portfolio

EDUCATION

University of California San Diego

La Jolla, CA

B.S. Electrical Engineering – Machine Learning and Control

Expected Dec 2026

RESEARCH INTERESTS

Robotic active matter and collective intelligence; nonlinear dynamics and emergent behavior in multi-agent systems; decentralized and learning-based control.

RESEARCH EXPERIENCE

UC San Diego — Research Assistant | PI: Dr. Tzer Han Tan

Nov 2024 – Present

Three Phases of Odd Robotic Active Matter

- Built and implemented custom MASBots and supporting experimental setups to study collective dynamics in nonreciprocal robotic active matter systems.
- Set up and calibrated experiments, and conducted systematic data recording for controlled multi-agent trials.
- Performed trajectory extraction, measurement, and data analysis to characterize MASBot interactions and collective behaviors.
- Modified and improved existing analysis code to process experimental data and examine dynamical patterns in the system.
- Contributed to data analysis and experimental validation for an ongoing research manuscript (see Publications).

Gear-Bot Collective Dynamics and Phase Transitions

Collaboration with Dr. Vishal P. Patil

- Lead a project investigating gear-driven robotic particles under spherical confinement, with emphasis on particle interactions, collective behavior, and mechanically induced phase transitions.
- Designed and fabricated the gear bots and experimental apparatus, developed testing protocols, and conducted systematic multi-agent experiments.
- Perform trajectory extraction and quantitative data analysis to characterize interaction mechanisms, emergent states, and transition behavior.

Data-Driven Modeling of Phase Separation in Robotic Active Matter

Collaboration with Prof. Jörn Dunkel

- Conduct experiments on phase-separating robotic active matter and analyze the resulting collective dynamics and spatial organization.
- Generate and analyze experimental trajectory data to support the development and validation of data-driven models for predicting larger-cluster behavior.

Learning Progress and Violation-of-Expectation (LEAP_VOE) Study

- Conduct infant research sessions using eye-tracking paradigms to investigate gaze behavior and information-seeking across violation-of-expectation and visual learning tasks.
- Recruit participating families, coordinate appointments, communicate with parents and community partners, and complete pre- and post-session procedures.
- Perform offline behavioral coding and analyze experimental data to identify patterns in infants' visual attention and learning behavior.

PUBLICATIONS & MANUSCRIPTS

* Authors contributed equally † Corresponding authors

Fan Bo*, Shiqi Liu*, **Zenghong He**, Wyatt Joyce, Gregor Leech, Kiet Tran, Keilan Ramirez, Nicholas Boechler, Nicholas Gravish, Hongbo Zhao, Tzer Han Tan[†]. Three phases of odd robotic active matter. Submitted.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB, Verilog, x86 Assembly

Modeling & Control: Linear control systems, optimization, nonlinear dynamics, data-driven modeling

Experimental & Hardware: Robotic prototyping, experimental system setup and calibration, motion tracking and trajectory analysis, circuit design, LTspice, Blender

Languages: English, Mandarin, Cantonese

TEACHING EXPERIENCE

De Anza College — Senior Tutor (Mathematics, Physics, Chemistry, Statistics) Apr 2023 – Jun 2024

- Provided individual and group instruction in quantitative subjects, supporting undergraduate students in problem-solving and conceptual understanding.
- Mentored and supervised junior tutors, coordinating instructional support across multiple STEM disciplines.